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10 CFR 50.73

GNRO2022-00006

February 2, 2022

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2021-005-00, Oscillation Power Range Monitors (OPRMs) Technical Specification Noncompliance.

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report 2021-005-00, Oscillation Power Range Monitors (OPRMs) Setpoint Error Causes Technical Specification Noncompliance. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), as an operation or condition prohibited by the plant's Technical Specification.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact Jeff Hardy, Regulatory Assurance Manager at 802-380-5124.

Sincerely,

JAH/fas

Attachments: Licensee Event Report 2021-005-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Attachment
Licensee Event Report 2021-005-00

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023
LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name	2. Docket Number	3. Page
Grand Gulf Nuclear Station, Unit 1	05000416	1 of 3

4. Title
Oscillation Power Range Monitors (OPRMs) Setpoint Error Causes Technical Specification Noncompliance

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name N/A	Docket Number 05000 N/A
12	06	2021	2021	005	00	02	02	2022	Facility Name N/A	Docket Number 05000 N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include Area Code)
Jeff Hardy, Manager Regulatory Assurance	(802)-380-5124

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ **Yes** (If yes, complete 15. Expected Submission Date)		N/A	N/A	N/A

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 21, 2021, while operating in MODE 1 at approximately 100 percent power, Grand Gulf Nuclear Station identified that the Detect and Suppress Solution – Confirmation Density (DSS-CD) settings for the Oscillation Power Range Monitors (OPRM) had not been updated to the current licensing basis as required per NEDC-33075P Rev 7 Licensing Topical Report General Electric Hitachi (GEH) Boiling Water Reactor Detect and suppress Solution – Confirmation Density. As a result, the Oscillation Power Range Monitors (OPRM) upscale setting requirements specified in Technical Specification (TS) Table 3.3.1.1-1, Reactor Protection System Instrumentation, function 2, Average Power Range Monitors (APRM), Sub-Function "f" OPRM upscale limits were not fully met.

This event is reportable under 10 CFR 50.73(a)(2)(i)(B), any operation or condition prohibited by the plant's Technical Specifications.

The direct cause of this event was that the setpoints were not implemented correctly during Maximum Extended Load Line Limit Analysis Plus (MELLLA+) implementation. The direct cause was corrected by changing the Maximum Oscillation Period, T_{max} setpoint to 4.0 seconds on all four (APRM) channels. This corrected the OPRM setpoints for DSS-CD.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Grand Gulf Nuclear Station, Unit 1	05000-416	2021	- 005	- 00

NARRATIVE

Plant Conditions:
Grand Gulf Nuclear Station (GGNS) Unit 1 was operating at 100 percent power in MODE 1. There were no Structures, Systems, or Components that were inoperable that contributed to this event.

Event Description:
On June 21, 2021, while performing analysis for a future fuel load, GGNS identified that the Detect and Suppress Solution – Confirmation Density (DSS-CD) settings had not been updated to the current version of the General Electric Hitachi (GEH) recommended report. Additionally, the Instrument and Control (I&C) Calibration procedures were not updated for the Average Power Range Monitor (APRM) channels. Specifically, the Maximum Oscillation Period, maximum time period (T_{max}), setpoint was set at 3.5 seconds instead of 4.0 seconds as required in the licensing basis per NEDC-33075P, Rev 7 Licensing Topical Report GEH Boiling Water Reactor DSS-CD.

The settings for the maximum period installed in the plant were non-conservative with respect to the licensing basis. The Oscillation Power Range Monitors (OPRM) Upscale setting requirements specified in the Technical Specifications (TS) 3.3.1.1, Reactor Protection System (RPS) Instrumentation, Table 3.3.1.1-1, Reactor Protection System Instrumentation, function 2, Average Power Range Monitors, Sub-Function "f", OPRM Upscale limits were not fully met.

This event is reportable under 10 CFR 50.73(a)(2)(i)(B), as any operation or condition prohibited by the plant's Technical Specifications.

Safety Assessment:
The actual consequences for this event were very low. There was no radiological release from the primary containment as a result of this calibration error. There were no other actual consequences to safety of the general public, nuclear safety, industrial safety, or radiological safety. No Technical Specification Safety Limits were violated.

Event Cause(s):
The direct cause of this event was that the setpoints were not implemented correctly during Maximum Extended Load Line Limit Analysis Plus (MELLLA+) implementation.

Causal Factors 1 – The technical basis for the change in OPRM settings was not clearly communicated such that consequences of not executing the work were well understood.

Causal Factors 2 – GGNS-NE-16-00007, the engineering report for the June 17, 2016, OPRM SCRAM, was inappropriately revised in a way that contradicted the Power Range Neutron Monitor (PRNM) DSS-CD Settings Report causing confusion regarding the actual licensing basis. The report was also inappropriately used to state T_{max} = 3.5 seconds was acceptable, also contradicting the PRNM DSS-CD Settings Report.

Corrective Actions:
The direct cause was corrected by changing the T_{max} setpoint to 4.0 seconds on all four APRM channels. This corrected the OPRM setpoints for DSS-CD.

Reperform an extent of condition on MELLLA+ Implementation for settings/setpoint changes. Tracked by the Corrective Action Program.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Grand Gulf Nuclear Station, Unit 1	05000-416	2021	- 005	- 00

Revise GGNS-NE-16-00007 to delete the changes made by revision 2 for CR-GGN-2021-06434, CA-0010. Revision 3 of GGNS-NE-16-00007 evaluates the DSS-CD settings for T_{min} = 1.05 and T_{max} = 3.5 and PBDA Sp = 1.15. Converting back to Revision 1 for CR-GGN-2021-6434, CA-10. This action is complete

Previous Similar Events:

LER 2016-009-00 / LER 2016-009-01

Entry into Mode of Applicability with the Oscillation Power Range Monitor Upscale Settings Incorrectly Set. The direct cause of this event was the failure to ensure the required procedure changes were incorporated and performed prior to the unit entering the mode of applicability.

Prior corrective actions did not prevent recurrence because, the technical basis for the change was not clearly communicated. The settings were updated in an evaluation engineering change. This engineering change type is considered a document only change.